

2.1. Overview

The key difference between Fixed-Price projects and T&M ones is the fact that most of the risk is transferred to the vendor, where budget, timeline, deliverables, and, sometimes, a delivery date are defined up front. In this situation:

- The customer pays an agreed fixed amount irrelative of the actual cost incurred
- The biggest part of the agreed fixed amount is usually paid after the whole project is delivered
- Some contracts include penalties if not delivered on time.

The discovery phase is about defining scope, budget, resource plan, milestones and assumptions of the overall project in a bounded timeframe. Fixed-price projects require very detailed requirements specification to get fine grain and reliable costs and efforts estimate. Discovery is more worthy to get you into the customer business than to make a detailed spec. Adjusting rough estimate to solid estimate comes not after all risks are identified and technical decisions are done, but after you only got into some understanding of their business specifics and some of their data structures.

Checklist

Conception and initiation

Get all constraints about the project from your Account Manager/Pre-sales team:

- ☐ Is it a highly recommended to have a separate Fixed-Price contract for Discovery phase?
- ☐ Potential project start date and hard deadline for project delivery
- ☐ Get/formulate preliminary discovery scope to identify Key EPAM stakeholders in Practices
- ☐ Is it a separate project or part of a bigger program?
- ☐ Is it a new customer for EPAM?
- ☐ Staffing location restrictions
- ☐ Rates
- ☐ Additional services (consulting, L1/L2/L3 support, etc.)
- ☐ High-level budget for implementation
- ☐ Constraints collected from the customer (architecture, schedule, processes)

Pre-discovery initiatives - get as much information about the customer and expected solution as you can:

- ☐ Company background
- ☐ Business domain
- ☐ Company culture
- ☐ **Decision makers**
- ☐ **Key stakeholders**
- ☐ Internal processes (or their absence)
- ☐ **Business processes**, especially CR's approval/signing procedures
- ☐ Enterprise architecture where you need to place you solution in

Definition and planning

Build a discovery team, involving

- ☐ Validate if Presales team's engagement is required and engage if so
- ☐ Competency center or Practice
- ☐ SA/Technical Lead with domain knowledge
- ☐ SME or BA with domain knowledge
- ☐ Schedule communications sessions with discover team to finalize discovery plan

- ☐ Discuss possibility to extend involvement of discovery team members up to the Project Implementation phase (ensuring continuity with practice/location Resource Managers)

Develop a discovery plan which will include workshops planning and time to work on the product backlog and architectural documents

- ☐ Overview session, where business goals will be shared and explained to the whole team
- ☐ Architecture/Infrastructure
- ☐ Integrations
- ☐ Sessions for each functional area
- ☐ Project communication plan
- ☐ Set up all needed sessions with all the relevant stakeholders

Internal kick-off

- ☐ Get discovery team confirmation of feasibility of discovery deliverables in expected timeline and quality.

Get access to customer infrastructure

- ☐ Request access to the customer infrastructure, tools and source code repositories required to execute the discovery as soon as possible
- ☐ PII access

Launch

- ☐ Run the kick-off meeting with the customer (Discovery Kick off Template.pptx)
 - ☐ Handshake, introduce teams
 - ☐ Validate timeline, it's phases, major milestones and expected deliverables
 - ☐ Verify assumptions
 - ☐ Confirm main goals and expectations
 - ☐ Agree on governance
 - ☐ Agree on next steps
 - ☐ Follow up

Governance, Monitoring and Controlling

- ☐ Set up governance touch points (internal and external)
 - ☐ Report progress and roadblocks
- ☐ Bring to the customer intermediate results to make corrective actions of running the discovery
- ☐ Run iterative sessions with the customer to discuss intermediate results

Execution

- ☐ Solution architecture
 - ☐ Big picture of the system(s) landscape
 - ☐ High-level shape of the software architecture and how responsibilities are distributed across it
 - ☐ Technology choices and tech stack
- ☐ **Backlog, scope identification**
 - ☐ Run the session, collecting all the requirements and building the Product backlog - identify scope
 - ☐ the format of the requirements is clear and there is approval with a defined impact on timeline/cost
 - ☐ guidelines and standards are gathered, described
 - ☐ Understanding business scope
 - ☐ get access to customer's training environments of legacy systems and learn how to do basic business workflows themselves. If customers make these actions manually - learn how to do it in Excel or even on paper. **It dramatically reduces the probability of the 'scope creep', which is really not 'creep', but 'detailing'.**
 - ☐ **high-level vision**, with a success definition from the senior stakeholder's perspective

- ☐ List **Functional Requirements (EPICs, Features)** with **assumptions**
- ☐ Define **integration points** with **assumptions**
- ☐ Define and confirm with the customer general test strategy and quality attributes including acceptance criteria, DoD, quality gates (if possible)
- ☐ Check if any pre-defined release strategy is exist and should be followed
- ☐ List **Non Functional Requirements (NFRs)**
- ☐ List **out of scope functionality**
- ☐ Voice out and document **all the assumptions/dependencies and their impact you have agreed on during the Discovery (incl. architectures assumptions)**- it will go to SOW for implementation
- ☐ **Analyze, qualify and estimate all risks** - it will go to overall estimation
 - ☐ General and project-specific risks
 - ☐ Specifically clarify NFR risks (pre-go-live assessment by any auditor, huge data migrations from legacy systems, etc.)
- ☐ Define automated and manual business processes that can be used in the project
- ☐ **Define and document key dates for dependencies**
 - ☐ Define end-date for architecture changes (all further changes will impact timeline/budget)
 - ☐ Data setup/Data import/Data migration readiness date
 - ☐ 3rd parties delivery dates
 - ☐ Check if any release windows exist and should be followed
- ☐ Collect methods that should be used during the project, if any take place
 - ☐ formally described estimation method
 - ☐ formally described re-estimation process after requirements are approved
 - ☐ categorization and prioritization (or severity) of bugs, format, SLA's and triaging process
 - ☐ process of reporting environment issues and SLAs defined assigned to the customer
- ☐ All Deliverables are described and confirmed with the discovery team and if possible potential (prescreened) execution key team members

Info

PAY SPECIAL ATTENTION

If you haven't yet analyze the amount of effort to be conducted during the project or EPAM didn't actively participate in the pre-sale phase and didn't helped to form customer budget

Costs calculation (contingency included)

- ☐ **Resource plan.** During the planning of the team, workload don't forget to use focus-factor (non-production value activities that could be in place during Delivery). This factor significantly influence final schedule dates. It's good to have 30% for non-production activities.
- ☐ List **licenses** that should be purchased - it can impact project budget
- ☐ Check if **infrastructure** (for test and production environments purpose) costs required to consider and if so:
 - ☐ Estimate Infrastructure expected costs if possible through EPAM partnership with tools/services supplier
 - ☐ Define assumptions
- ☐ Create a sprint (period) based milestone project plan, which should have
 - ☐ Development start (and all further dates are relative to the development start date)
 - ☐ **Milestones**
 - ☐ Delivery dates for Themes/EPICs and/or if required for major Features
 - ☐ Delivery dates for all phases and also stages with the customer and 3rd parties (UAT, E2E, transition, etc)
 - ☐ Overall delivery date
- ☐ List estimates for the activities and phases
 - ☐ granular estimates provided by trusted technical subject matter experts. The estimates should be justified by technical and business assumptions against each functional and non-functional point
 - ☐ Try to challenge received estimations to get more detailed estimations. After a couple of times, you will find out the level of risk, that we should take into count and should list to discuss with the Account Manager.
 - ☐ You should merge the level of the risk into the final estimates. All the risks should be noted and not be as huge as possible.

- ☐ Overall risk value shouldn't be more than 30% of original estimates. In case your risk is more huge - details the requirements and continue estimations and decomposition.

Delivery team staffing

- ☐ Create an opportunity/convert an existing opportunity to a project in Staffing Desk
- ☐ Create a preliminary resource plan in Staffing Desk: roles, primary skills, workload
- ☐ Add positions description
- ☐ Initiate staffing process with WPM team and locations/practices

Finalization and demo of outcomes and proposal

- ☐ Do an internal dry-run engaging BU, Account/Delivery, Practices stakeholders based on recommended proposal content:
 - ☐ Executive summary
 - ☐ Clarified Customer Key Stakeholders and decision makers
 - ☐ Solution architecture
 - Big picture of the system(s) landscape
 - High-level shape of the software architecture and how responsibilities are distributed across it
 - Technology choices and tech stack
 - ☐ Requirements
 - Backlog with features descriptions, assumptions (**efforts limitations, boundaries**)
 - Integrations
 - NFRs
 - Out-of-scope items
 - ☐ **General and technical assumptions lists**
 - ☐ **Dependency register with impact and points of no return**
 - ☐ **Risk register**
 - ☐ Licenses
 - ☐ Resource Plan
 - ☐ Budget. **Don't forget to add the margin rate to the final estimates (if you haven't done it earlier)**
 - ☐ Project Schedule
 - ☐ Test strategy and Testing approach
 - ☐ Change request flow agreement (including approval gates, accountable representatives, templates, etc.)
- ☐ Demo a final proposal to the customer
- ☐ Draft SOW, check with AM if required

2.2. Duration

Discovery duration totally depends on the Project size. It can start from 2 weeks (minimal recommended duration for Fixed-Price Projects) and last up to up to 3-4 months for complex multi-integrations projects (up to 30% of overall project duration).

The customer needs to show back some value from the project within short period, they don't trust vendors to run long discoveries, as vendors don't know their business and fail too often - risk of not having well-defined requirements and architecture after a long discovery is too high.

2.3. Useful Information

Artifacts & Examples

- Discovery Agenda (example)
- Discovery Template (example)
- Scope Baseline (example)
- Risk List (example)
- Standard Risk Categories and Risk Sources
- Risk Sources Categorization
- Milestone Plan

Tips & Tricks

 Tabs Page | title = Plan Discovery

Planning to minimize the chance of unexpected risks and dependencies

Going through already gathered requirements without the customer can be beneficial to find any missing information or any dependency that was not obvious during the workshops.

- During Workshop planning leave some time for internal syncs (1 day in the middle of the week) where you can carefully go through already gathered requirements and outline possible dependencies and risks

 Tabs Page | title = Study Customer

Defining the delivery approach

Knowledge of things like Business domain, Company culture, Key stakeholders and Internal processes (or their absence), End-to-end business process and Enterprise architecture is essential in order to be able to identify **dependencies**, predict **issues**, uncertainties, and amount of change requests, and be able to create a realistic plan. This knowledge will also help define the delivery approach that is the most suitable.

- Is there any chance to propose something other than FP? It won't hurt to try and convince the customer to switch to another type of contract if it's beneficial for EPAM too.

 Tabs Page | title = Run the Workshops

For the successful workshops:

- Plan, define who should join particular sessions and send an invites with the Agenda and expected information to be prepared beforehand
- If the customer representatives are not straightforward with sharing necessary information that can be converted into requirements - then prepare a presentation and drive a Q&A session

 Tabs Page | title = Identify Scope

Build your Backlog

Having a good scope breakdown is even more important than hunting for details. Each work breakdown item has to be INVEST, which means there is a business value associated with it and it can be independently delivered to production.

- Concentrate on the business goal
- Having a good scope breakdown is key - each backlog item has to be INVEST
- Build a backlog out of a scope breakdown
- Prioritize backlog, putting MVP items first, **ordered by business value and confirmed with the customer**
- Elaborate as much as you can on each backlog item
- List non-functional requirements

 Tabs Page | title = Identify Dependencies

Start managing risks & issues

When creating a list of dependencies, consider the following areas:

- Customer's internal processes, e.g. sign off / acceptance process, response time
- UX, design assets
- Third party systems and services (performance testing, security audit, etc.)
- Environments and infrastructure
- Access to key stakeholders and their availability
- List of dependencies from an initial list of risks and assumptions, based on which a roadmap and a contract will be built



Build your roadmap

If the statement of work has already been signed, with the budget set in stone and scope not, it is still recommended to validate the budget and assess the situation your project is in. It will then help concentrating on MVP and help minimize risks and losses.

Info

Backlog of user stories can be put against a timeline in the same way a traditional plan is created, only in this case, instead of having 2 major blocks for development and testing, it will be built of backlog items ordered by their MVP, priority and external dependencies, if any. Each backlog item, when done, would be developed, tested and deployed into an environment, which makes the product an incremental release candidate.

- Estimate backlog items. If somebody sees translating t-shirt sizes or story points into tangible budget as a taboo, there are plenty more estimation approaches and describing them would require a separate article.
- [When the delivery date is fixed, include in the SOW only MVP backlog items. The remaining scope can be flexible \(either CRs or remaining backlog\)](#)



Avoid common mistakes

There are a lot of issues, mistakes and challenges that thousands of managers face during the estimation, so you can study their mistakes and don't let it happen in your project.

Here is the list of typical issues during the estimation phase - Estimation & Planning

